

Solutions for Practice Problems for Week 3

Operating and Systems Programming 2022/23

1. A function pointer is is a pointer to a C function. They are used to pass functions as arguments.
2. The first piece of code declares `foo` to be a function which expects an integer and returns a pointer to an integer. The second piece of code declares `foo` to be a pointer to a function which expects an integer and returns an integer.
3.
 - Have 4 arguments
argument 0 is `./program`
argument 1 is `arg1`
argument 2 is `arg2`
argument 3 is `arg3`
 - Have 5 arguments
argument 0 is `./program`
argument 1 is `First`
argument 2 is `argument`
argument 3 is `Second`
argument 4 is `argument`
 - `./program "First argument" "Second argument"`
4. Void pointers are used when the type of the pointer is unspecified. They are a poor replacement for the missing generic types in C.
5. Different types have different sizes. The compiler needs to know the correct size, eg for memory allocation or correct interpretation of the location the pointer points to. In the following program

```
struct type1 {  
    int x;  
};  
  
struct type2 {  
    int x;  
    int y;  
}
```

```

void foo (void *args) {
    struct type2 *p;

    p = (struct type2 *) args;
    printf ("The value of y is %d\n", p->y);
}

```

```

int main (int argc, char **argv) {
    struct type1 *p;

    p = malloc(sizeof(struct type1));
    p->x = 4;
    foo((void *) p);
    return 0;
}

```

the effects are undefined - no memory has been allocated for the field y.

6. A Makefile lists all dependencies for a set of programs and contains instructions how to build large programs.
7. The commands will be

```

cc -Wall -Werror -g -c input.c
cc -Wall -Werror -g -c program.c
cc -Wall -Werror -g -o program input.o output.o program.o

```